

Introduction

Non-inferiority (NI) trials test whether a new treatment is not worse than an active comparator by more than a pre-specified margin Δ . Margin selection is the most consequential and scrutinised part of NI trial design: too loose and a genuinely inferior treatment is approved; too tight and the study is infeasible.

Prerequisites

Hypothesis testing; active-controlled trials; historical-control data.

Theory

Two common approaches: - **Fixed-margin (synthesis) method**: margin Δ is chosen based on the historical effect of the active comparator vs placebo, typically preserving 50-75 % of that effect. Example: if active reduces mortality by 10 % vs placebo, Δ might be set at 5 %. - **Clinical margin**: a clinically judged threshold of practical importance, independent of historical data.

Both require regulatory justification; FDA and EMA typically require the synthesis approach with supporting clinical judgement.

Assumptions

Historical active-vs-placebo effect is consistent and generalisable to the current trial population; assay sensitivity (ability to detect a difference if truly present) is preserved.

R Implementation

```
# Synthesis-method margin calculation
# Historical effect: placebo-controlled active gives risk reduction 10% (95% CI 7%-13%)
# Conservative estimate: lower bound 7%
preservation <- 0.5 # preserve at least 50% of effect
margin_synthesis <- 0.07 * (1 - preservation)
cat("Synthesis-based NI margin:", margin_synthesis, "\n")

# Sample size for an NI trial with continuous outcome
# Expected true difference 0; margin 0.25 SD; alpha=0.025; power=0.80
library(pwr)
pwr.t.test(d = 0.25, sig.level = 0.025, power = 0.80,
           type = "two.sample", alternative = "greater")
```

Output & Results

Synthesis-based margin ($0.035 = 3.5\%$ preserving 50 % of historical effect); sample-size calculation gives ~252/arm for a 0.25 SD margin.

Interpretation

“The non-inferiority margin was prospectively set at 3.5 percentage points, preserving at least 50 % of the historical benefit of the active comparator (7 % lower bound of historical effect). Sample size was 504 based on 80 % power to rule out a margin with one-sided alpha 0.025.”

Practical Tips

- Margin selection must be pre-specified, regulator-reviewed, and clinically justified.
- Run both ITT and per-protocol analyses; NI requires consistency.

- Beware “biocreep”: repeated non-inferiority approvals without placebo anchoring drift from effective therapy.
- Assay sensitivity is hard to demonstrate without a placebo arm; three-arm trials (NI + placebo) are ideal but often unethical.
- Non-inferiority claims should report the effect estimate and CI, not just “ $p < 0.025$ ”.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale